Rehab Center Experience — Presentation Script

585.617 Rehabilitation Engineering | Module 10 | target 9:45, hard limit 10:00

Timing

Slide	Segment	Start	Length
1	Title	0:00	0:15
2	Project topic	0:15	1:00
3	The center	1:15	0:45
4	Patient observation	2:00	0:40
5	What the observation changed	2:40	1:00
6	Technology overview	3:40	0:40
7	Relevance to my project	4:20	0:55
8	Care provider	5:15	0:35
9	Interview themes 1	5:50	0:45
10	Interview themes 2	6:35	0:50
11	Impact on preliminary design	7:25	1:10
12	Gaps and next steps	8:35	0:40

About 1,270 spoken words. At a measured 140 words a minute that finishes near **9:15**, which leaves 45 seconds of margin. If you overrun, cut the third point on slide 5 and the last sentence on slide 7 rather than rushing slide 11.

Slide 1 — Title (0:00)

Hi, I'm Vance Van Volkenburgh. This is my Module 10 rehab center experience. I visited the Kennedy Krieger Institute in Baltimore on November tenth, and spent my time on their inpatient Neurobehavioral Unit.

Slide 2 — Project Topic (0:15)

Thirty seconds on the project first, so the rest makes sense.

I'm designing an interface that lets a manual wheelchair user operate handheld powered yard tools — trimmers, edgers, hedge trimmers, blowers — on their own yard. It's scoped as an interface and mounting problem. Not a new tool, not a new wheelchair.

[point at the figure and walk the three numbered callouts]

What makes it hard is on the right. Three constraints are coupled. One: the tool pushes back, and the chair is the only thing available to react that force, so tool torque eats into a stability envelope that's already narrow. Two: the working envelope runs from ground level to hedge height, from a seat that doesn't move. Three: propelling a manual chair is two-handed, so holding a tool occupies the limbs mobility needs.

Any one is solvable. The coupling isn't. I went in with no design selected, on purpose.

Slide 3 — The Center (1:15)

Kennedy Krieger is a pediatric institute in Baltimore — brain, spinal cord and musculoskeletal disorders, up to age twenty-one. The inpatient rehab unit is twenty-five beds, CARF and Joint Commission accredited, with several hours a day of occupational, physical and speech therapy.

But I want to be precise about where I was, because it shapes everything after. I was on the Neurobehavioral Unit — a specialist service for children and adolescents with developmental disabilities and severe problem behavior. The model is applied behavior analysis, and admissions run weeks to months.

[pause here]

So: not a mobility service. I'll come back to that.

Slide 4 — Patient Observation (2:00)

I observed a behavioral treatment session, not a physical or occupational therapy session.

What stuck with me wasn't the clinical content. It was how the session was built.

Behavior was recorded continuously, in real time, as it happened. Staffing was high, and one person was there purely for safety. And the environment was part of the treatment — surfaces, fixtures and equipment all chosen for what they prevent.

Patient privacy limited what I could see, and I didn't record or photograph anything.

Slide 5 — What the Observation Changed (2:40)

Three things went straight into my design thinking.

First, measurement is part of the intervention, not paperwork. Nothing was judged by impression. That's a direct criticism of my PDS — I've been writing aims like “reduce effort.” It needs measurable acceptance criteria instead: task time, number of attempts, setup time.

Second, staff presence is a design variable. A task needing a second person is a different product. My version is whether the user can deploy and stow the tool without a second pair of hands, and I hadn't treated that as a requirement.

Third, and this changed my mind: safety is designed into the environment before it's trained into the person. I'd been treating stability as something the user manages by technique. Better to design so the unsafe state is hard to reach.

Slide 6 — Technology Overview (3:40)

On technology there's a clear split.

Used constantly, and deliberately low-tech: visual schedules, token boards, timers, picture cards, protective equipment, padded surfaces.

Also constant but sophisticated: real-time behavioral data capture on tablets, and video monitoring. Used selectively: communication devices, including speech-generating ones.

What I didn't expect is that the most sophisticated element isn't hardware. It's the functional analysis protocol itself — the hardware supporting it is simple on purpose. And durability dominates: equipment has to survive deliberate destruction, not just wear.

Slide 7 — Relevance to My Project (4:20)

Now the honest part. I saw nothing related to my project. No wheelchair-mounted tooling, no outdoor task equipment, very few manual wheelchair users.

[pause here]

That absence is structural. The unit treats behavior. It isn't a mobility service, and its population isn't mine.

But the constraints transfer, and they're harder than a rehab gym would impose. Abuse tolerance against deliberate destruction, not incidental wear. Cleanability as a requirement. No ligature points, no entrapment, nothing that detaches or gets thrown. And engagement as an engineering requirement — if the user disengages, the device has failed, however good the mechanics.

If my interface can meet that bar, residential yard use is a far easier environment. These constraints aren't from my domain, but they're stricter than it.

Slide 8 — Care Provider (5:15)

I interviewed Daniel Dobin, a Clinical Assistant Two on the Neurobehavioral Unit. Eighteen months on the unit, and he now trains incoming clinical assistants alongside direct patient care.

I chose that role deliberately. He works with the equipment daily rather than prescribing it. He trains the people who use it, so he knows which parts of a protocol survive a real unit. And he sees the same devices across many patients.

[say this plainly and move on]

He asked that his photograph not be used, so there isn't one.

Slide 9 — Interview Themes, One of Two (5:50)

Two themes. First, constraints and priorities.

Safety screening comes before function. Ligature risk, pinch points, parts that detach or can be thrown — pass or fail, not trade-offs. Cleanability sits with safety, not convenience.

[point at the ranking chart]

Asked to rank needs, this is what came out. Essential: safe with limited supervision, durability, cleanability. Important: setup speed, and holding the user's attention. Lower: cost.

[pause here]

Cost ranking lowest was the surprise. I'd assumed affordability was the binding constraint. It isn't. That ranking is what I carry into the Kano diagram for the Preliminary Design Review.

Slide 10 — Interview Themes, Two of Two (6:35)

Second theme: how things fail, and what he made of my directions.

Devices fail by breakage under deliberate force, lost parts, setups too slow to be worth starting, and patients disengaging. Mechanical failure wasn't the most common mode — not where I'd have put my money.

[point at the three panels]

I showed him the three directions. Carry the tool and its reaction on the chair. Route the force into the ground. Or reduce the force at source. He preferred the third: fewer attachments means fewer failure points and faster setup.

I also asked for a referral to a therapist or seating specialist, since stability and funding sit outside his role.

Slide 11 — Impact on My Preliminary Design (7:25)

[slow down - this is the slide the rubric names]

So what actually changes.

One: quick-release moves from nice-to-have to essential. Clearest single message of the visit, and it's safety-driven.

Two: abuse tolerance and cleanability enter the PDS as requirements, not finishing details.

Three: abandonment becomes a design criterion. If setup is slow enough that the user stops bothering, the device has failed. Setup time gets a number.

Four: acceptance criteria become measurable — attempts, elapsed time, setup time.

Five: scope confirmed, source corrected. The framing holds, but the quantitative inputs have to come from an adult mobility clinician, not this unit.

[point at the task analysis]

The diagram is where that lands — a task analysis of trimming a lawn edge from a manual chair, using the Module 8 method. Three steps are red: propelling over turf, deploying the tool from stow, and holding it against the reaction force. That's where my design has to act, and the visit is why deploying from stow is on the list at all.

Slide 12 — Gaps and Next Steps (8:35)

Two things I couldn't get. No client interview — a pediatric inpatient behavioral unit, where privacy and the clinical setting made it inappropriate, so the user voice is still missing. And no quantitative mobility numbers; tipping margins and reimbursement sit outside a direct-care behavioral role.

Next steps. A therapist or seating specialist for the numbers. Interview adult manual wheelchair users, the population I'm designing for. Carry the needs ranking into a Kano diagram. And use the safety and durability constraints to screen concepts.

Thank you.

Delivery notes

Pace. About 1,270 spoken words. At 140 words a minute that lands near 9:15. If you speak quickly when nervous, slow down deliberately on slides 5, 7 and 11 — those three carry the content grade.

Do not read the slides. The rubric penalises it by name. The slides carry the headline; the script carries the reasoning. Where a bullet and a sentence overlap, say the sentence.

Three places to pause. After “not a mobility service” on slide 3. After “I saw nothing related to my project” on slide 7. And after “cost ranking lowest was the surprise” on slide 9. Each one lands better with a beat after it.

Rehearse once against a clock before recording. If you overrun, cut the third point on slide 5 and the final sentence on slide 7 rather than rushing slide 11.